

# 사이클별 지표 패널

## M02Ch104

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM\_PE / \*\_proxy / pattern\_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R\_ohmic은  $\sqrt{t}$  외삽 proxy (0.1s 단일값  $\neq$  순수 옴).

### ## 트렌드 요약 — M02Ch104

- 유효 지표: 48개 · aging 방향 일치 17 · 반대 1

#### ### 하락 트렌드 (상위)

- LAM 곡선 proxy: early=-2.983 → late=-16.28 ( $\Delta$ =-13.3, -5.282%/100cyc) → decreasing · context
- LLI 곡선 proxy: early=-0.6179 → late=-2.925 ( $\Delta$ =-2.307, -1.265%/100cyc) → decreasing · context
- 방전 플래토 폭: early=16.87 → late=14.8 ( $\Delta$ =-2.067, -1.075Q-units/100cyc) → decreasing · matches\_aging
- R $\Omega$  성장률 /100cyc: early=0.7957 → late=0.4668 ( $\Delta$ =-0.3289, -0.3314m $\Omega$ /100cyc/100cyc) → decreasing · context
- 방전 플래토  $\Delta V$ : early=-0.03799 → late=-0.08104 ( $\Delta$ =-0.04305, -0.01868V/100cyc) → decreasing · context

#### ### 상승 트렌드 (상위)

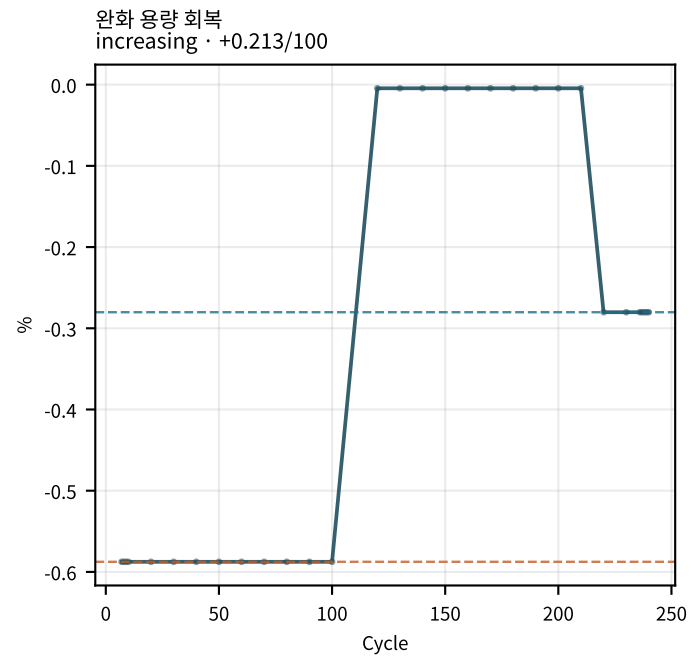
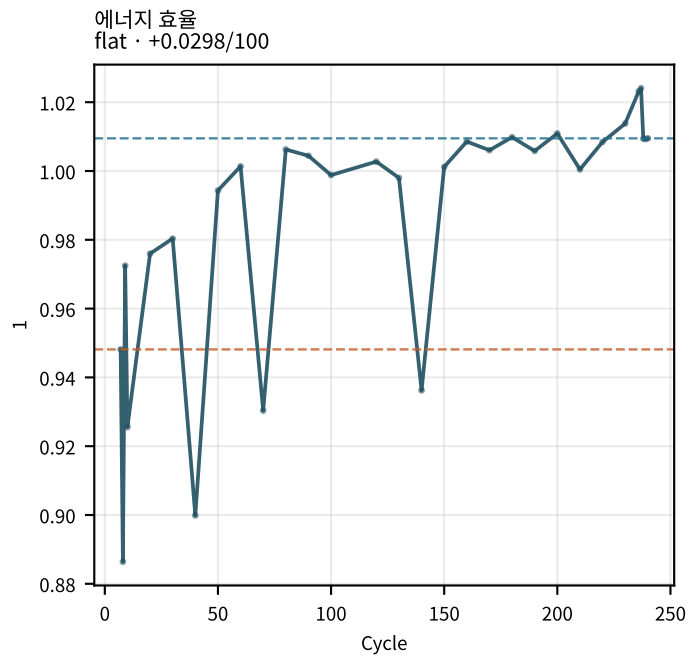
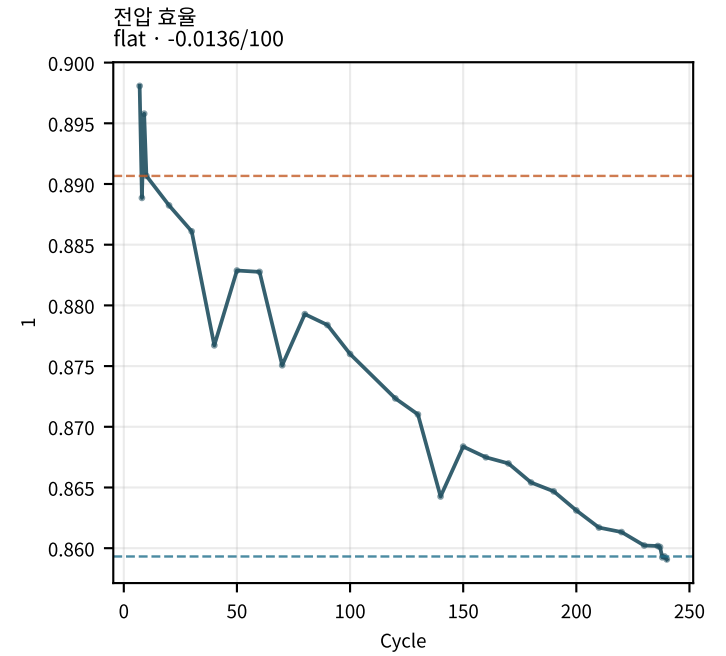
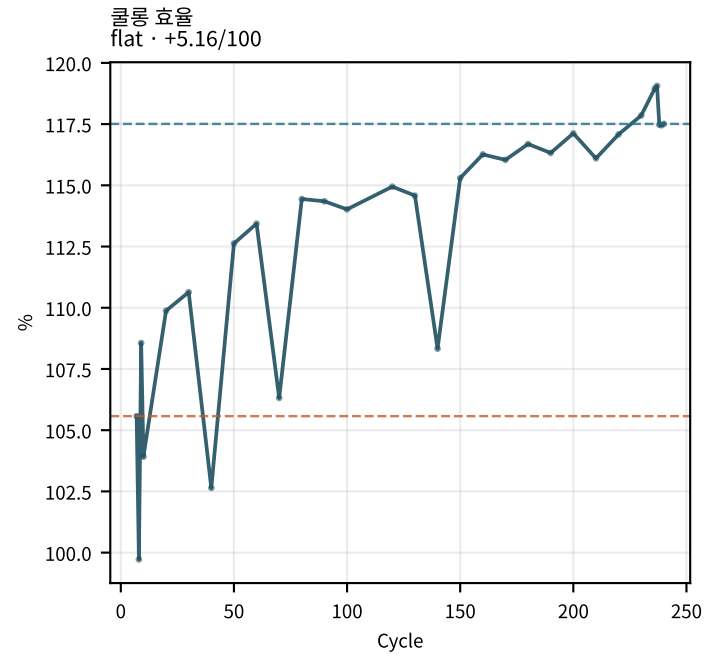
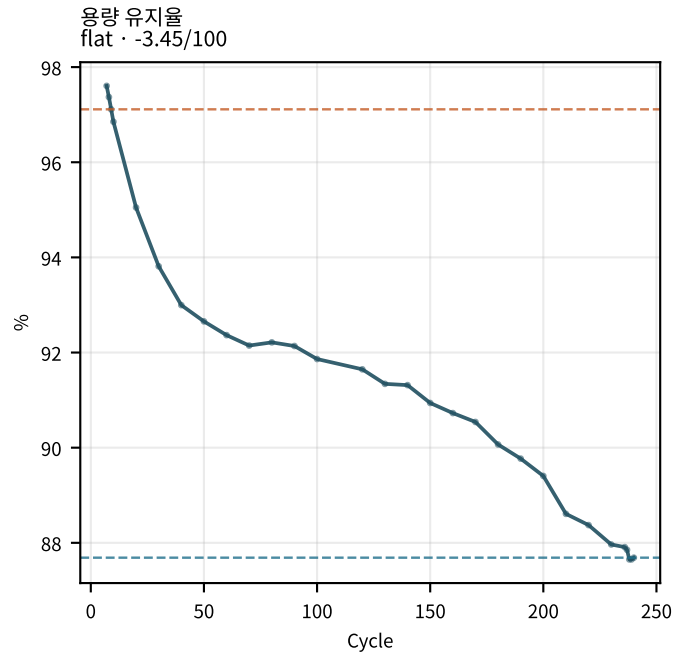
- 옴 저항 (SOC50): early=1.394 → late=2.374 ( $\Delta$ =+0.9802, +0.4981m $\Omega$ /100cyc) → increasing · matches\_aging
- 기계/화학 비: early=1.924 → late=2.519 ( $\Delta$ =+0.5948, +0.35921/100cyc) → increasing · matches\_aging
- 완화 용량 회복: early=-0.5876 → late=-0.2802 ( $\Delta$ =+0.3073, +0.2126%/100cyc) → increasing · matches\_aging
- 계면 R 패턴: early=0.4752 → late=0.8706 ( $\Delta$ =+0.3954, +0.14840-1/100cyc) → increasing · matches\_aging
- PE activity 패턴: early=0.6051 → late=0.9263 ( $\Delta$ =+0.3212, +0.14480-1/100cyc) → increasing · matches\_aging

CycleDiag : plot\_cycle\_metrics\_panel.py

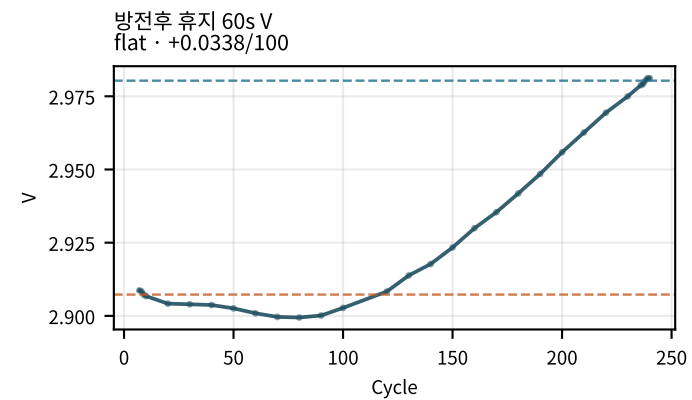
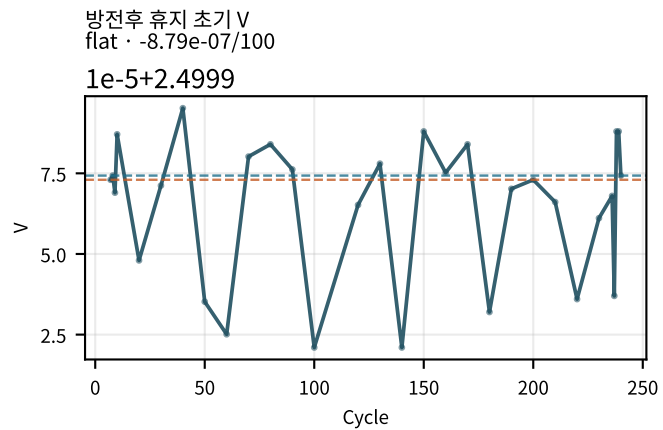
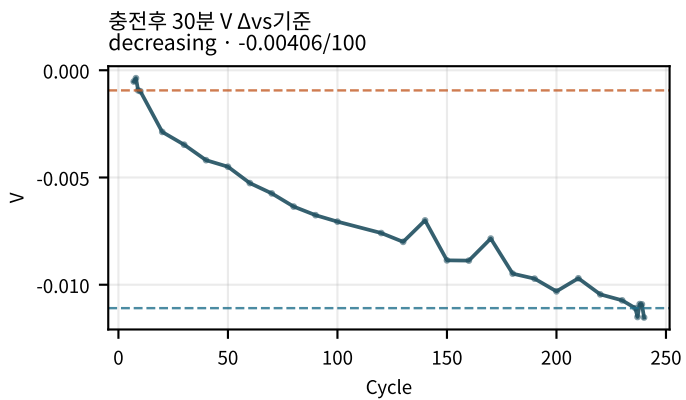
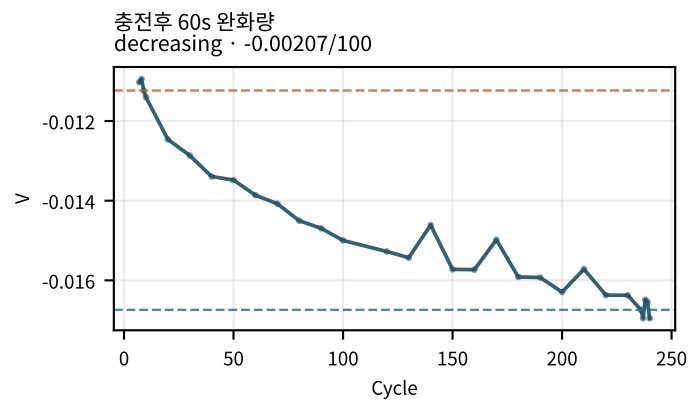
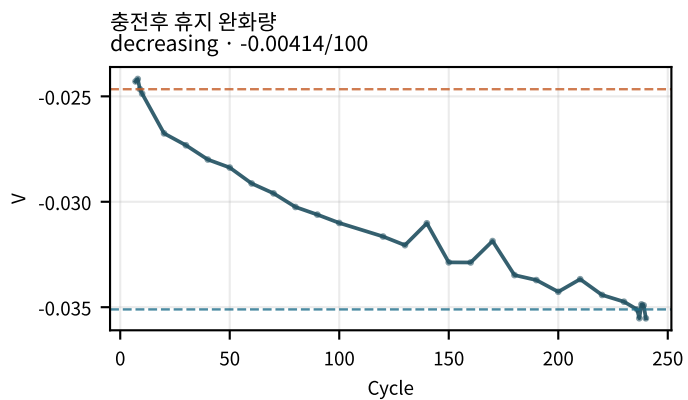
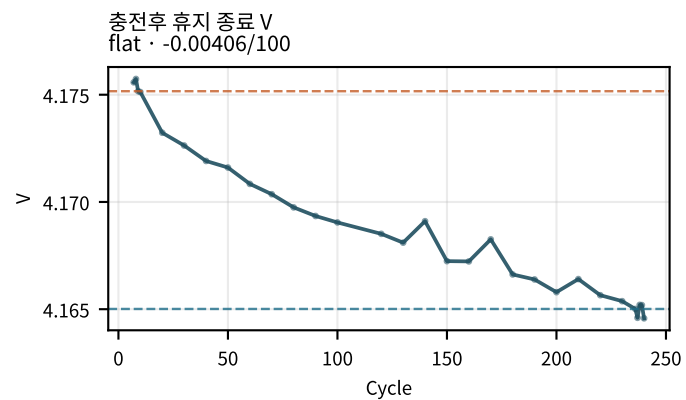
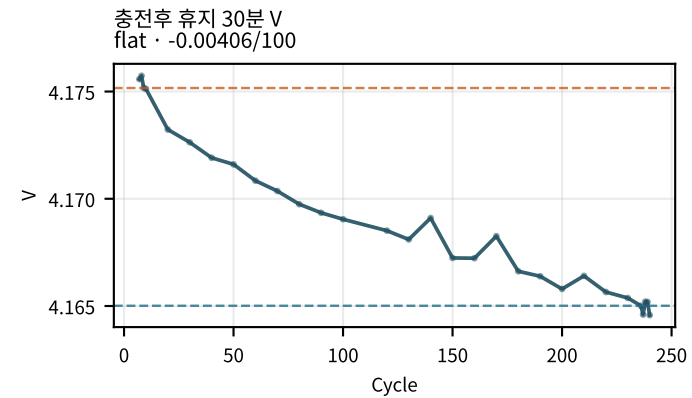
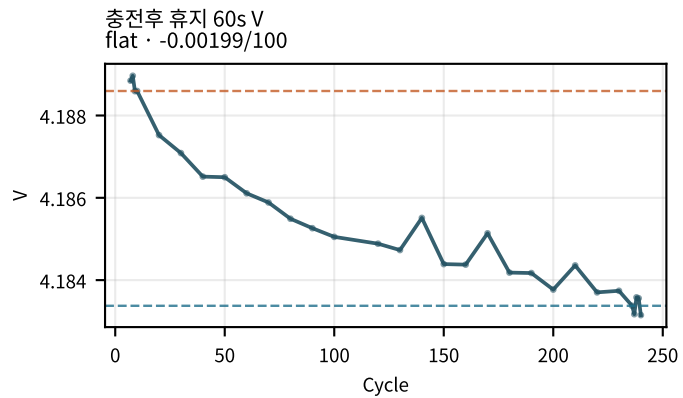
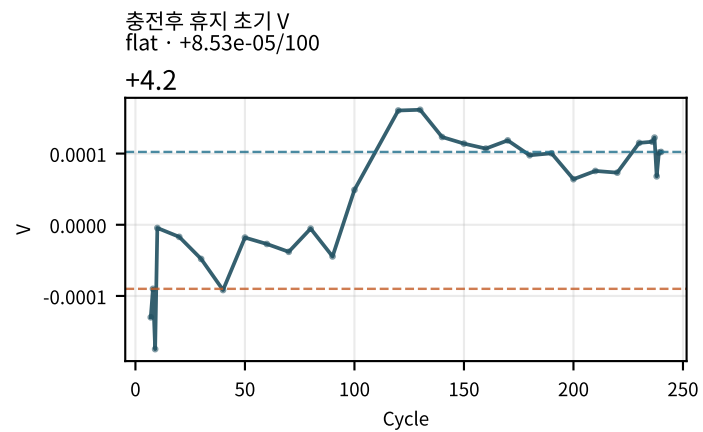
#### ### 기대 aging과 반대

- 저SOC 히스테리시스: early=0.07466 → late=0.04496 ( $\Delta$ =-0.0297, -0.01201V/100cyc) → decreasing · opposite\_aging

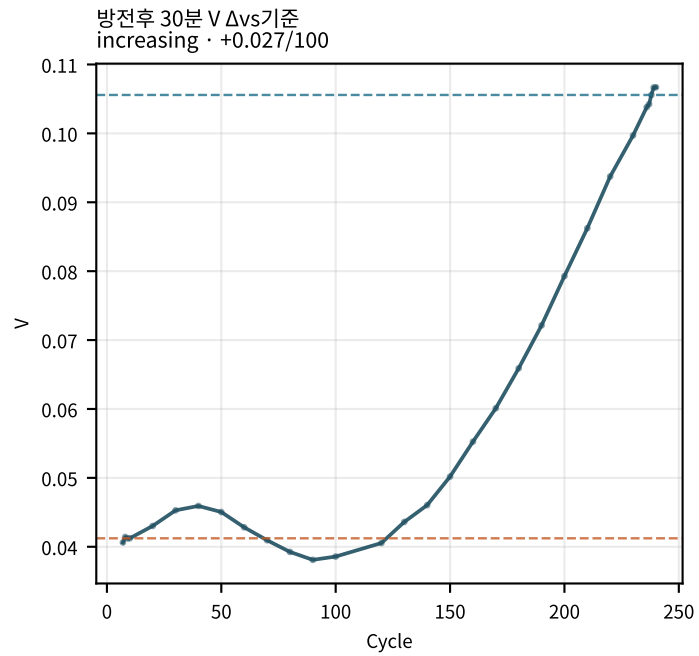
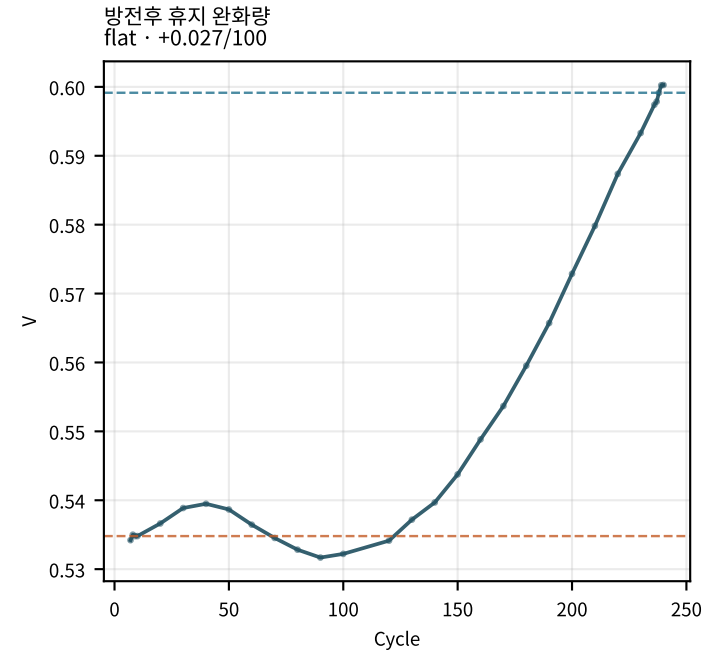
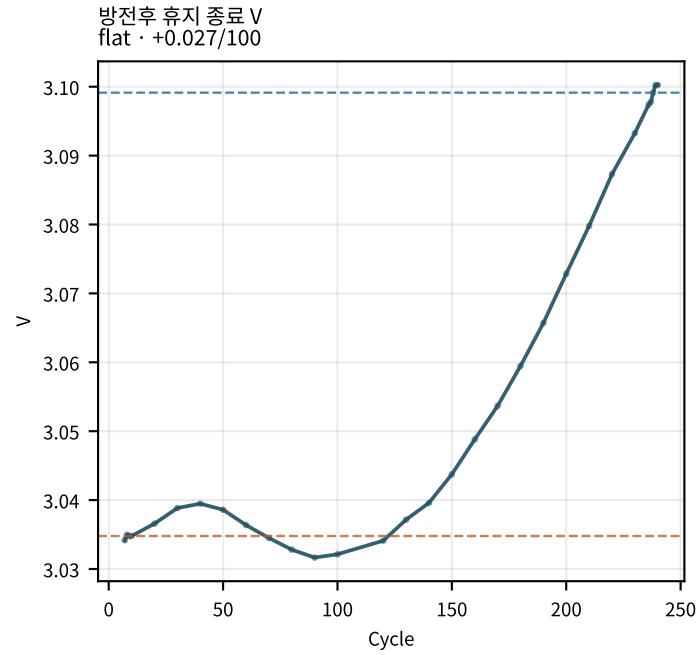
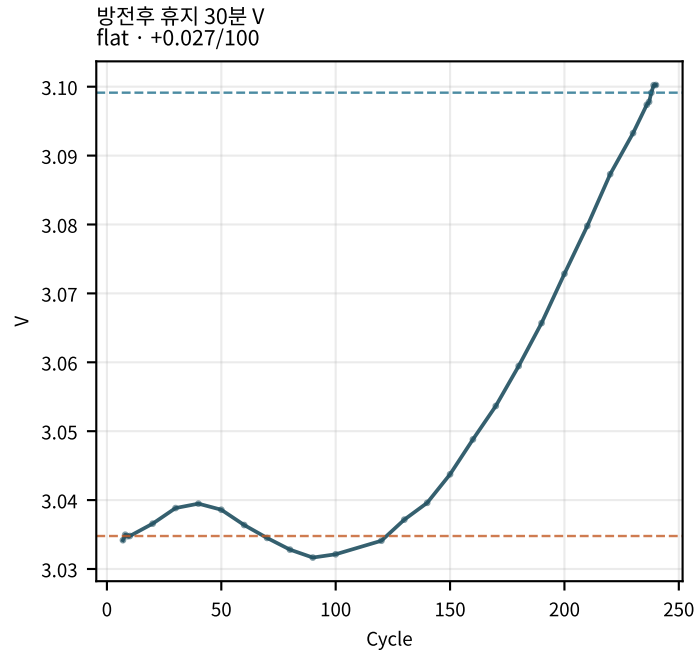
# M02Ch104 · 용량 · 효율



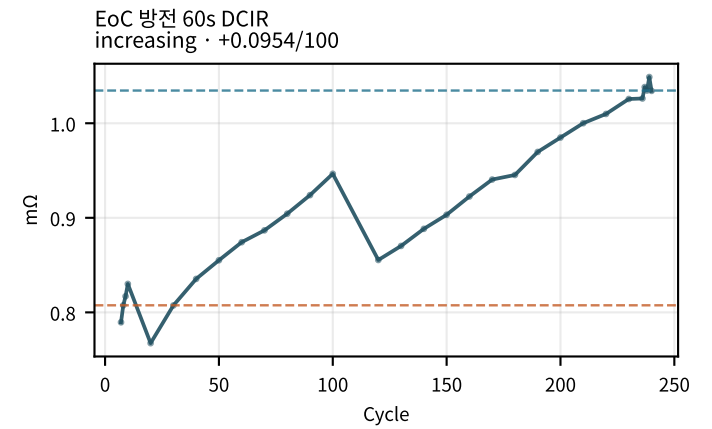
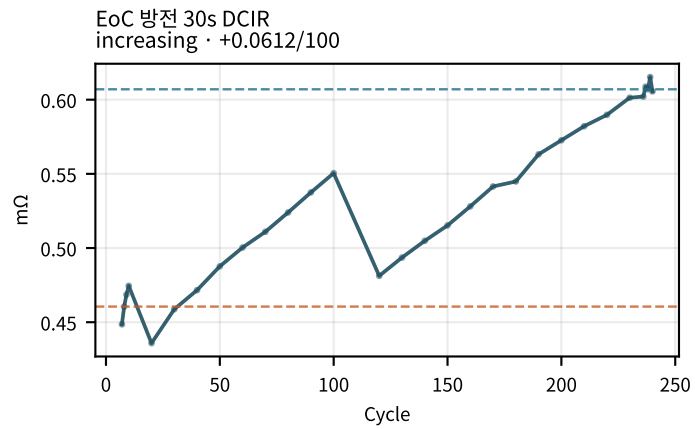
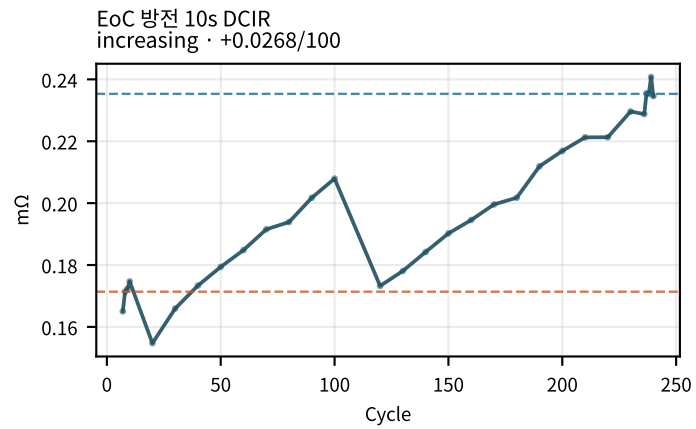
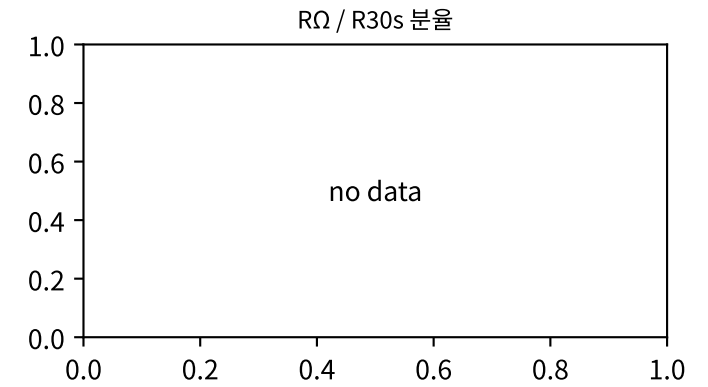
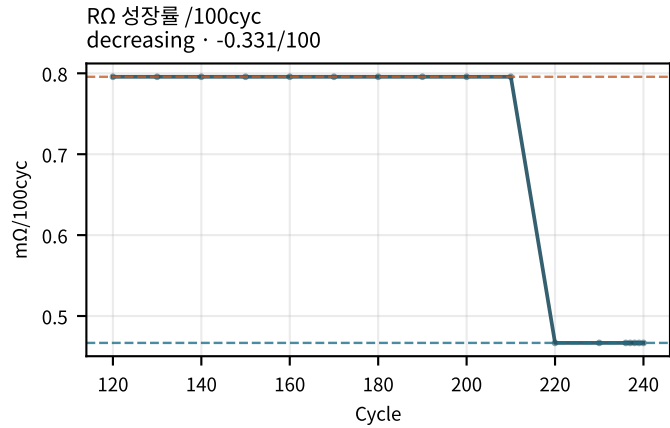
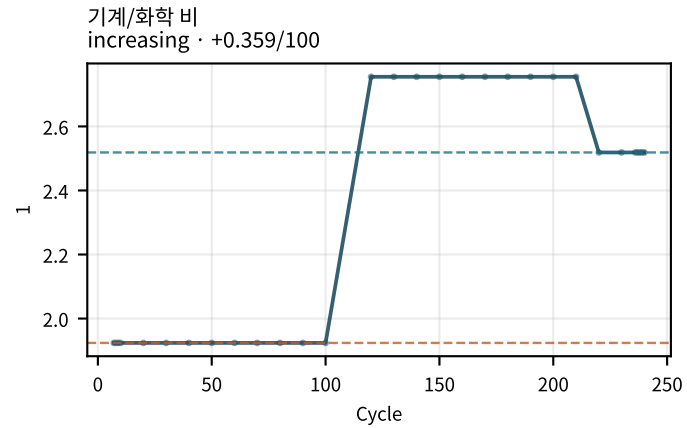
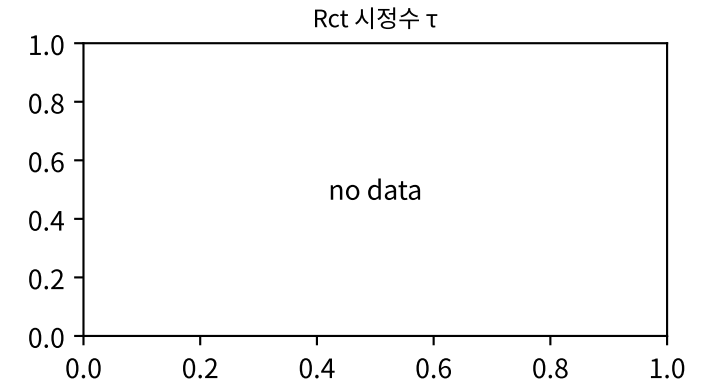
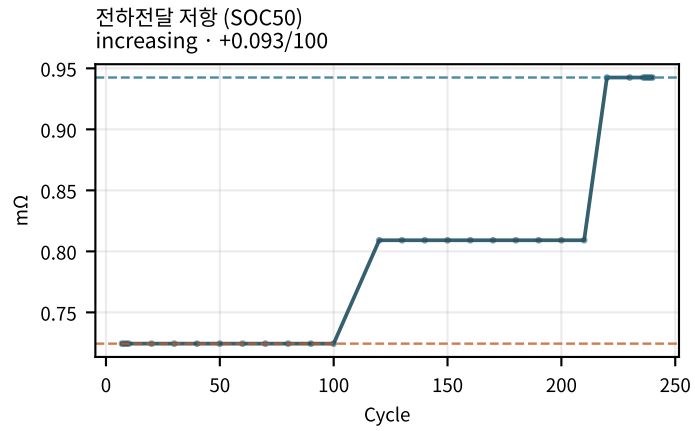
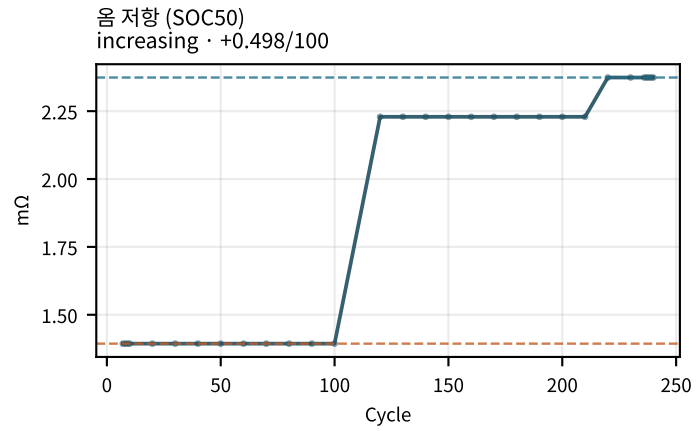
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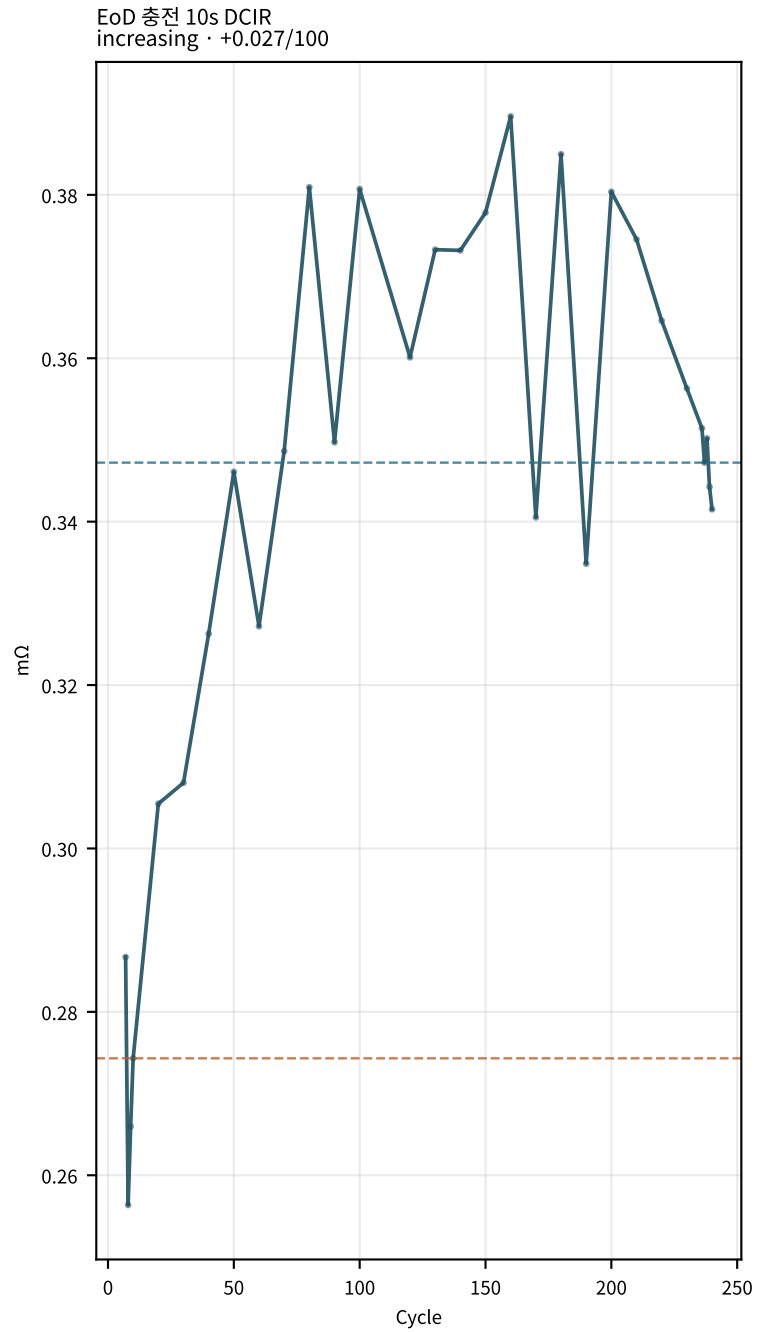
# M02Ch104 · 휴지 전압 (충전/방전 후)



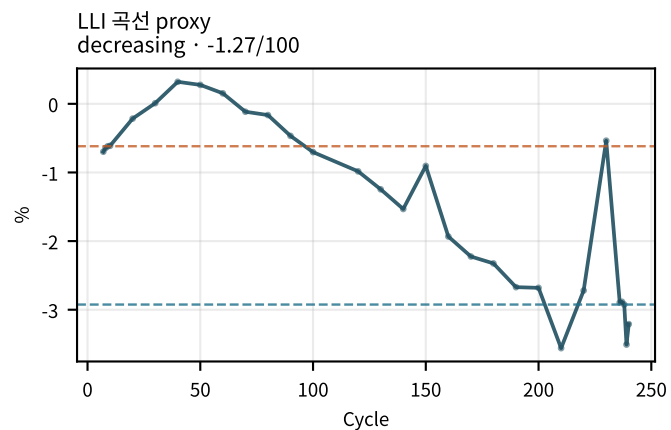
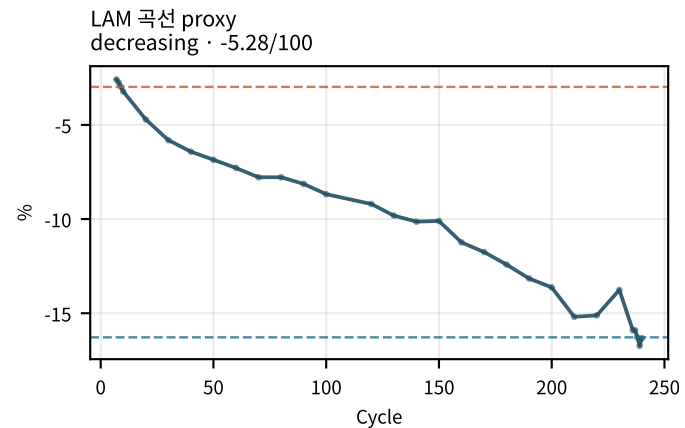
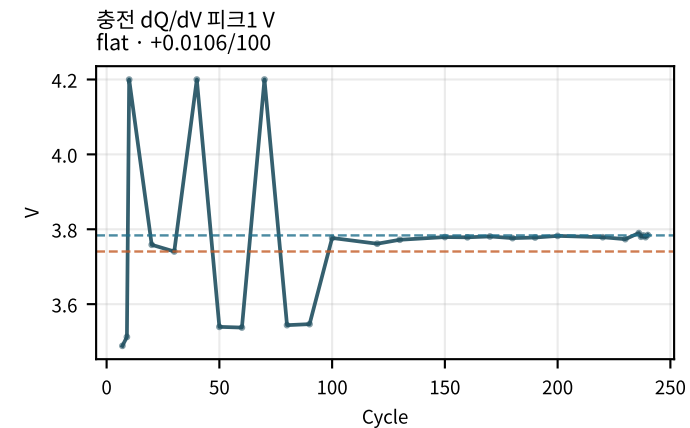
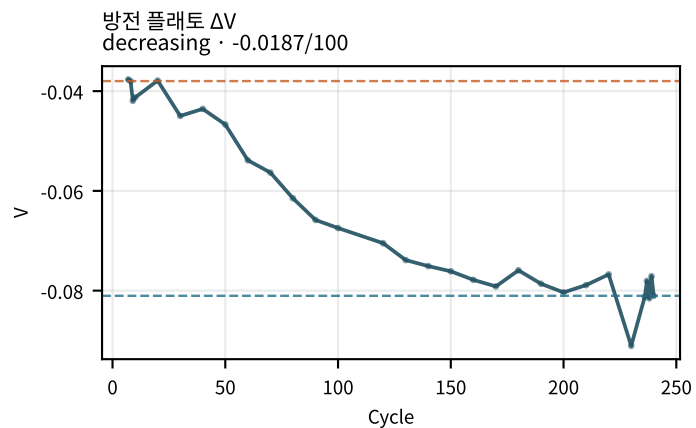
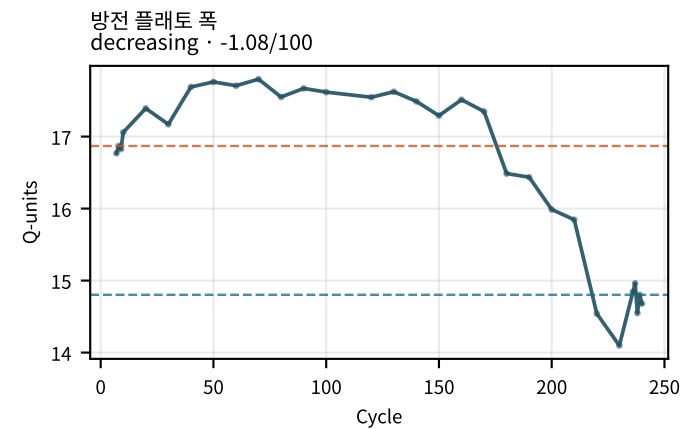
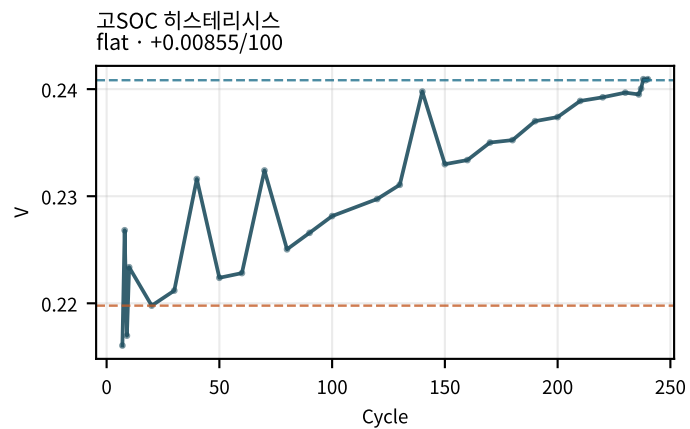
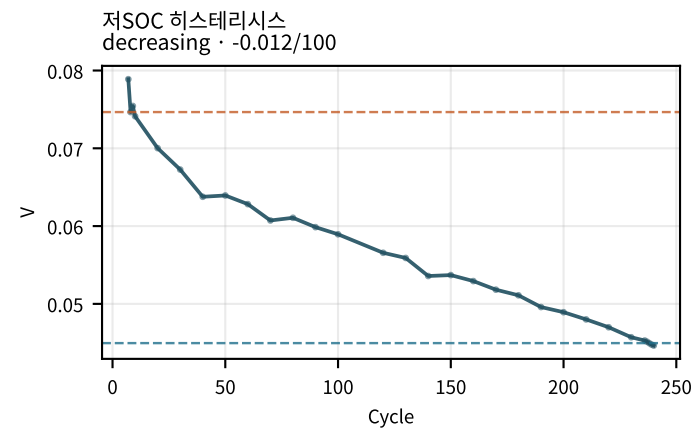
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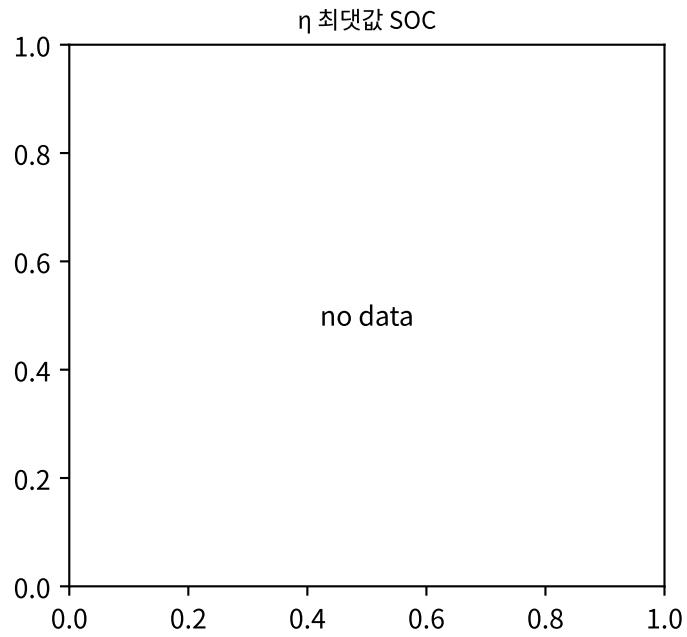
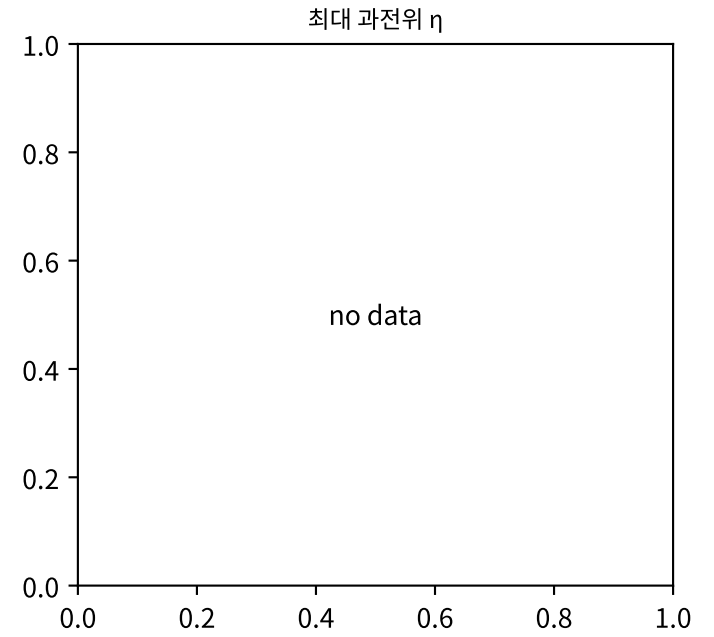
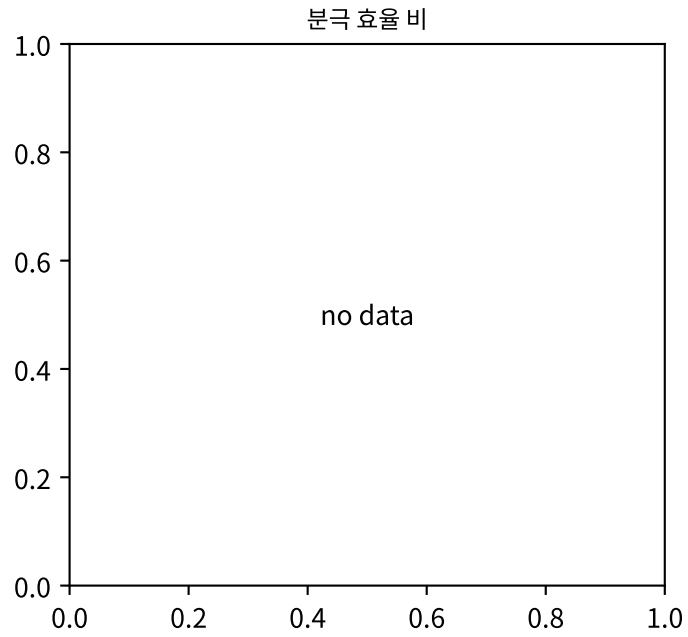
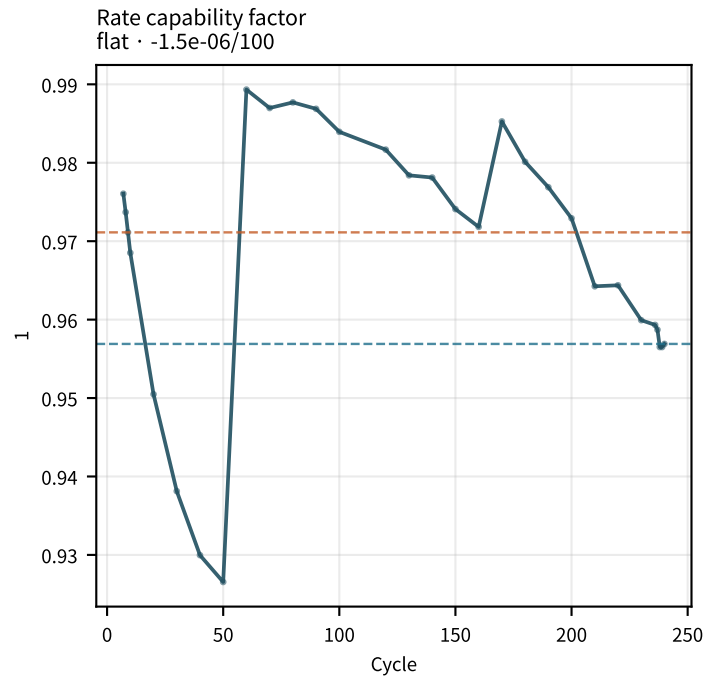
# M02Ch104 · 저항 · 분해



# M02Ch104 · 곡선 형상 · dQ/dV

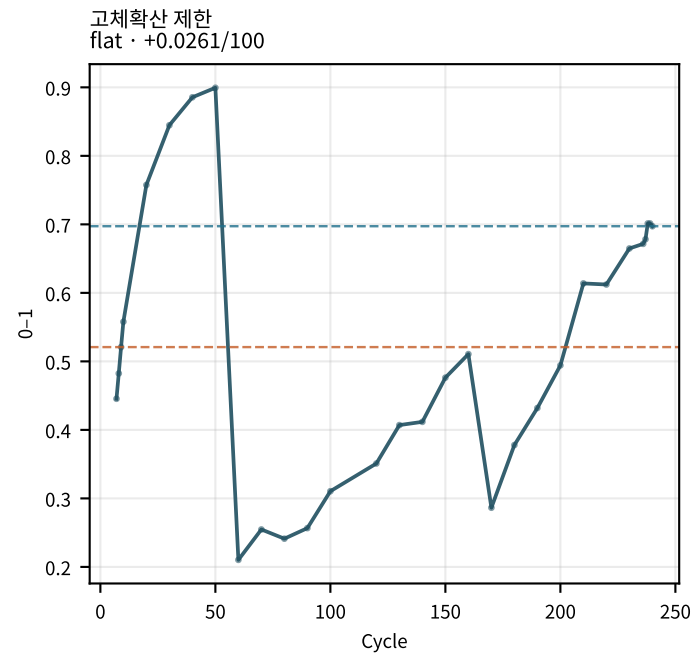
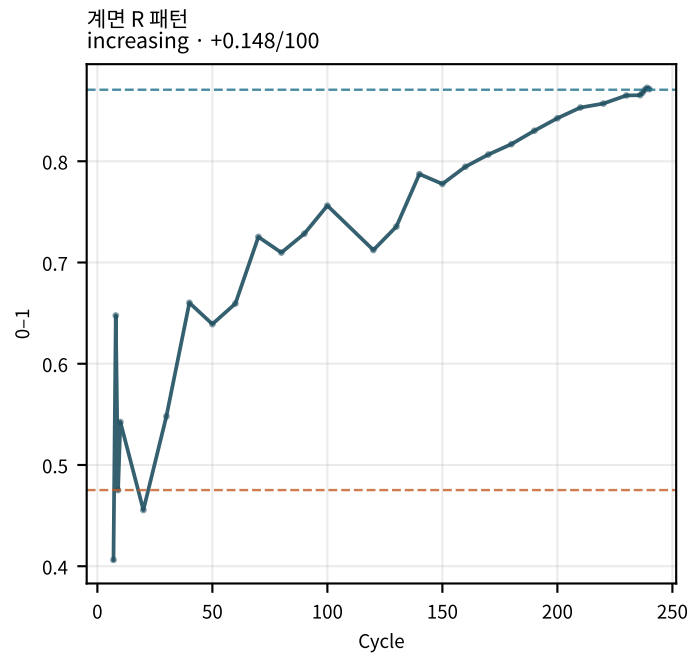
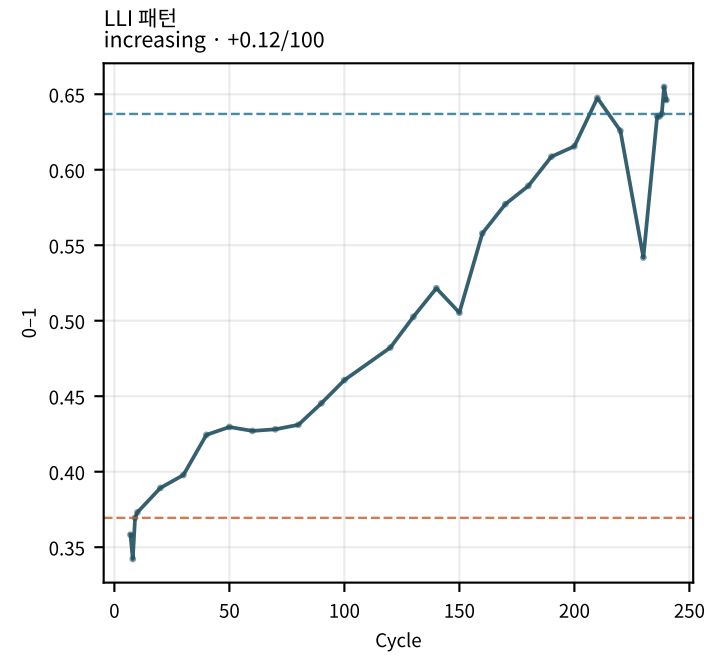
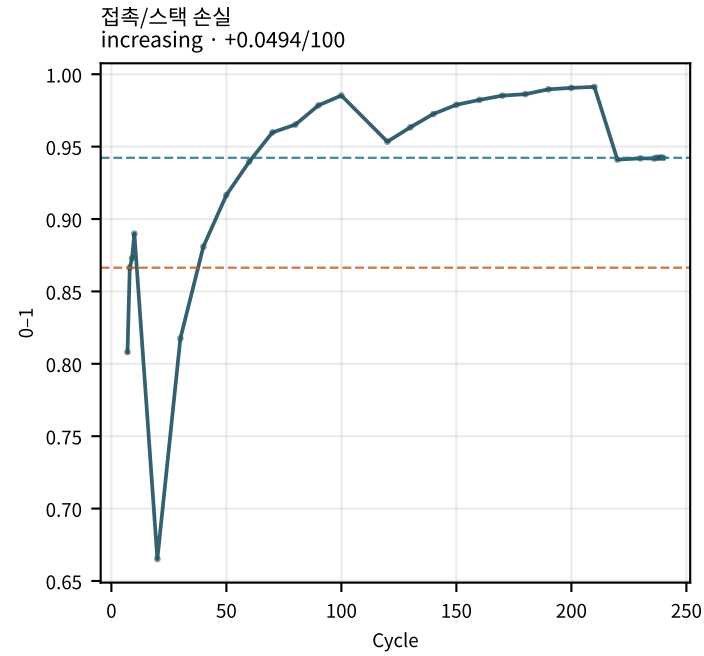
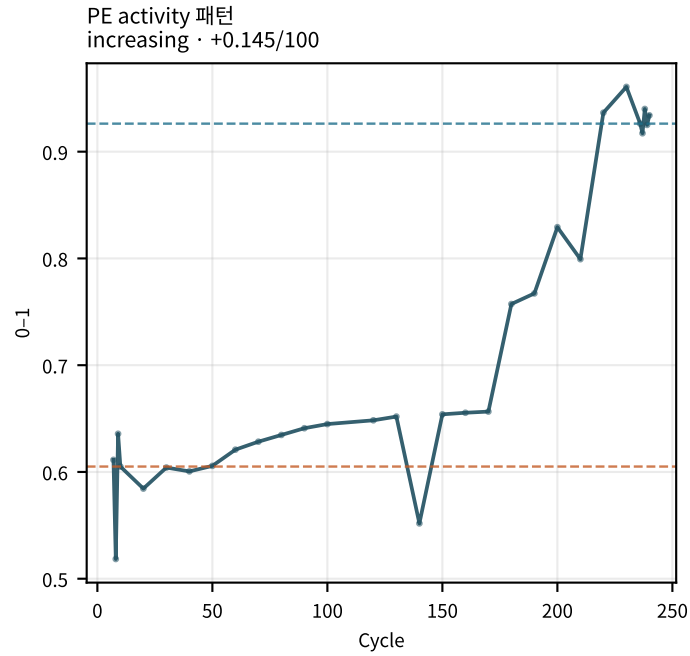


# M02Ch104 · 수송 · rate

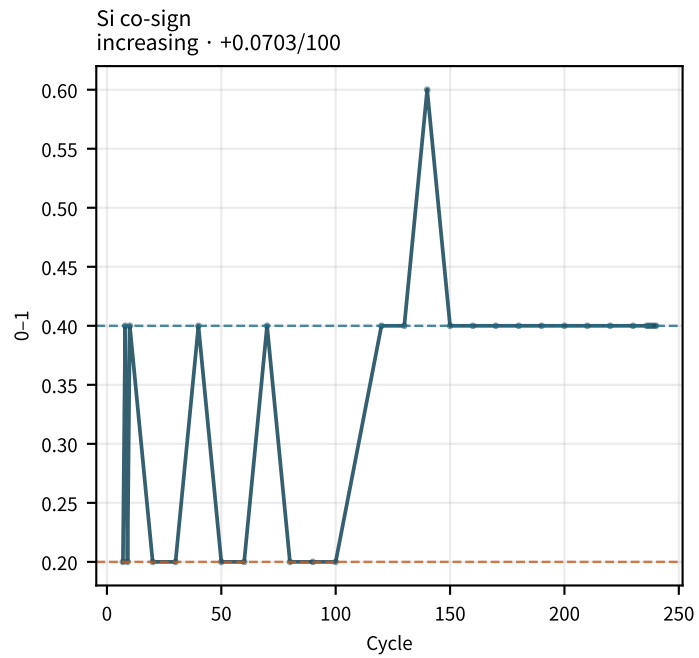
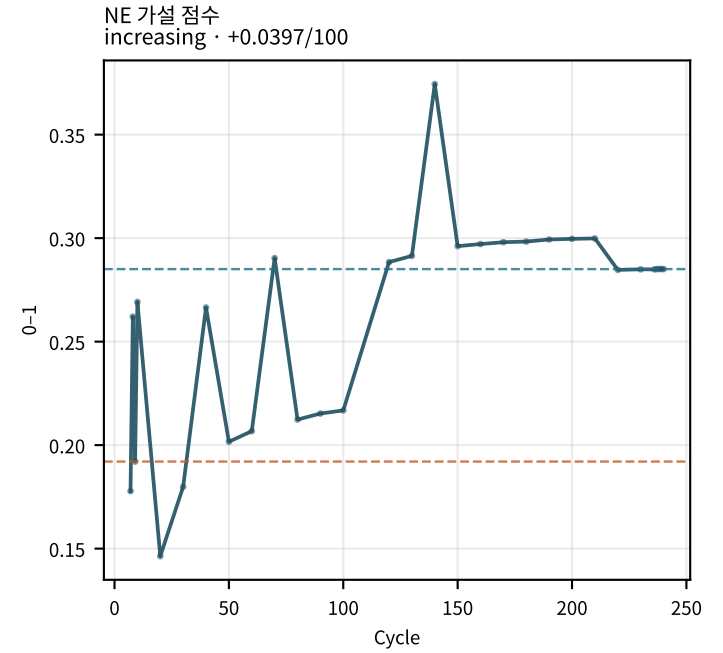
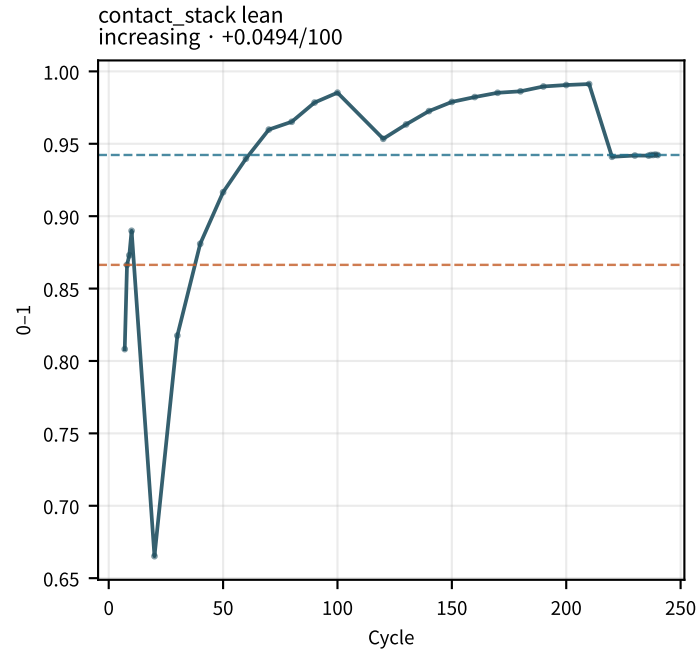
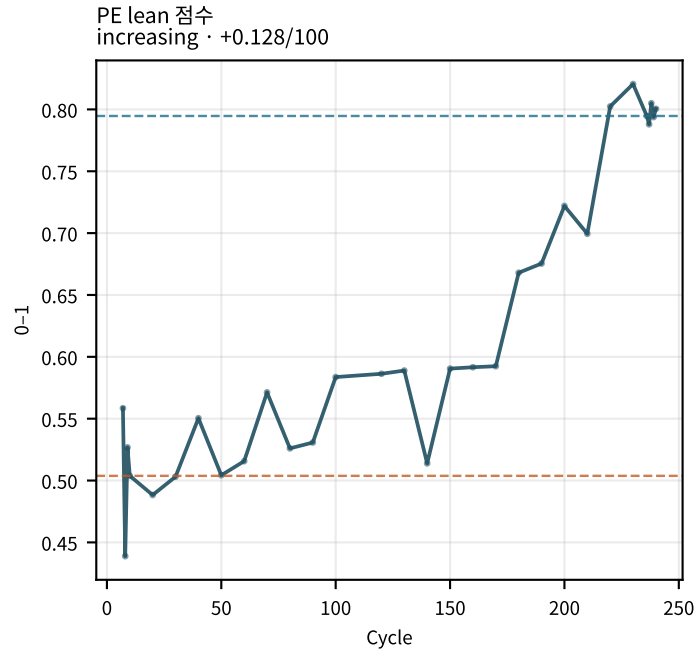




# M02Ch104 · 열화 패턴 점수



# M02Ch104 · 전극 lean 가설



## 지표 설명 · 트렌드 (M02Ch104)

### 용량 유지율 (SoHQ)

기준 대비 방전용량 유지율. fade/knee의 1차 관측량. | 계산:  $Q\_dchg(cycle) / Q\_baseline \times 100$  (routine 0.5C).

트렌드: 용량 유지율: early=97.11 → late=87.69 ( $\Delta=-9.425$ , -3.446%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

### 쿨롱 효율 (CE)

충전 대비 방전용량 비. LLI·부반응 힌트. | 계산:  $Q\_dchg / Q\_chg \times 100$  (이상치 >102% 또는 <85%는 진단에서 제외).

트렌드: 쿨롱 효율: early=105.6 → late=117.5 ( $\Delta=+11.94$ , +5.162%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

### 전압 효율 (VE)

에너지 효율의 전압 측. 저항·분극 증가 시 하락. | 계산:  $E\_dchg / E\_chg$  (Wh).

트렌드: 전압 효율: early=0.8907 → late=0.8593 ( $\Delta=-0.03135$ , -0.013621/100cyc) → flat · stable | aging 기대=decrease | vs=stable

### 에너지 효율 (EE)

충·방전 에너지 비. | 계산:  $E\_dchg / E\_chg$ .

트렌드: 에너지 효율: early=0.9481 → late=1.009 ( $\Delta=+0.06136$ , +0.029831/100cyc) → flat · stable | aging 기대=decrease | vs=stable

### 완화 용량 회복 (Q\_relax\_pct)

DCIR/RPT 전후 휴지로 회복되는 용량 분율. Si chemo-mech co-sign. | 계산: DCIR 블록 전후 Q 차이 / Q × 100, routine에 forward-fill.

트렌드: 완화 용량 회복: early=-0.5876 → late=-0.2802 ( $\Delta=+0.3073$ , +0.2126%/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### 충전후 휴지 초기 V (EoC\_restV\_init)

충전(EoC) 직후 휴지 시작 전압. | 계산: charge 종료 후 rest step 첫 샘플 전압.

트렌드: 충전후 휴지 초기 V: early=4.2 → late=4.2 ( $\Delta=+0.0001923$ , +8.526e-05V/100cyc) → flat · context | aging 기대=either | vs=context

## 지표 설명 · 트렌드 (M02Ch104)

### 충전후 휴지 60s V (EoC\_restV\_60s)

충전 후 휴지 60초 시점 전압. | 계산: rest step\_time  $\approx$  60 s 보간.

트렌드: 충전후 휴지 60s V: early=4.189  $\rightarrow$  late=4.183 ( $\Delta$ =-0.005219, -0.001986V/100cyc)  $\rightarrow$  flat · context | aging  
기대=either | vs=context

### 충전후 휴지 30분 V (EoC\_restV\_30m)

충전 후 휴지 30분 시점 전압. OCV에 가까운 EoC rest. | 계산: rest step\_time  $\approx$  1800 s 보간.

트렌드: 충전후 휴지 30분 V: early=4.175  $\rightarrow$  late=4.165 ( $\Delta$ =-0.01015, -0.004059V/100cyc)  $\rightarrow$  flat · context | aging  
기대=either | vs=context

### 충전후 휴지 종료 V (EoC\_restV\_end)

충전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 충전후 휴지 종료 V: early=4.175  $\rightarrow$  late=4.165 ( $\Delta$ =-0.01015, -0.004059V/100cyc)  $\rightarrow$  flat · context | aging  
기대=either | vs=context

### 충전후 휴지 완화량 (EoC\_restV\_relax)

휴지 동안 전압 변화 (end – init). 분극 완화. | 계산: EoC\_restV\_end – EoC\_restV\_init.

트렌드: 충전후 휴지 완화량: early=-0.02466  $\rightarrow$  late=-0.03511 ( $\Delta$ =-0.01044, -0.004144V/100cyc)  $\rightarrow$  decreasing · context |  
aging 기대=either | vs=context

### 충전후 60s 완화량 (EoC\_restV\_relax\_60s)

휴지 첫 60초 완화량. | 계산: EoC\_restV\_60s – EoC\_restV\_init.

트렌드: 충전후 60s 완화량: early=-0.01123  $\rightarrow$  late=-0.01674 ( $\Delta$ =-0.005506, -0.002071V/100cyc)  $\rightarrow$  decreasing · context |  
aging 기대=either | vs=context

### 충전후 30분 V $\Delta$ vs기준 (delta\_EoC\_restV\_30m)

기준 사이클 대비 충전후 30분 rest V 이동 (slippage/LLI 힌트). | 계산: EoC\_restV\_30m(cycle) – EoC\_restV\_30m(baseline).

트렌드: 충전후 30분 V  $\Delta$ vs기준: early=-0.000939  $\rightarrow$  late=-0.01109 ( $\Delta$ =-0.01015, -0.004059V/100cyc)  $\rightarrow$  decreasing · context  
| aging 기대=either | vs=context

## 지표 설명 · 트렌드 (M02Ch104)

### 방전후 휴지 초기 V (EoD\_restV\_init)

방전(EoD) 직후 휴지 시작 전압. | 계산: discharge 종료 후 rest step 첫 샘플.

트렌드: 방전후 휴지 초기 V: early=2.5 → late=2.5 ( $\Delta=+1.3\text{e-}06$ ,  $-8.795\text{e-}07\text{V}/100\text{cyc}$ ) → flat · context | aging 기대=either | vs=context

### 방전후 휴지 60s V (EoD\_restV\_60s)

방전 후 휴지 60초 전압. | 계산: rest  $\approx$  60 s 보간.

트렌드: 방전후 휴지 60s V: early=2.907 → late=2.98 ( $\Delta=+0.07306$ ,  $+0.0338\text{V}/100\text{cyc}$ ) → flat · context | aging 기대=either | vs=context

### 방전후 휴지 30분 V (EoD\_restV\_30m)

방전 후 휴지 30분 전압. OCV에 가까운 EoD rest. | 계산: rest  $\approx$  1800 s 보간.

트렌드: 방전후 휴지 30분 V: early=3.035 → late=3.099 ( $\Delta=+0.06435$ ,  $+0.02702\text{V}/100\text{cyc}$ ) → flat · context | aging 기대=either | vs=context

### 방전후 휴지 종료 V (EoD\_restV\_end)

방전 후 휴지 스텝 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 방전후 휴지 종료 V: early=3.035 → late=3.099 ( $\Delta=+0.06435$ ,  $+0.02702\text{V}/100\text{cyc}$ ) → flat · context | aging 기대=either | vs=context

### 방전후 휴지 완화량 (EoD\_restV\_relax)

방전 후 휴지 완화 (end – init). | 계산: EoD\_restV\_end – EoD\_restV\_init.

트렌드: 방전후 휴지 완화량: early=0.5348 → late=0.5991 ( $\Delta=+0.06435$ ,  $+0.02702\text{V}/100\text{cyc}$ ) → flat · context | aging 기대=either | vs=context

### 방전후 30분 V $\Delta$ vs기준 (delta\_EoD\_restV\_30m)

기준 대비 방전후 30분 rest V 이동. | 계산: EoD\_restV\_30m(cycle) – baseline.

트렌드: 방전후 30분 V  $\Delta$ vs기준: early=0.04123 → late=0.1056 ( $\Delta=+0.06435$ ,  $+0.02702\text{V}/100\text{cyc}$ ) → increasing · context | aging 기대=either | vs=context

## 지표 설명 · 트렌드 (M02Ch104)

### 옴 저항 (SOC50) (R\_ohmic\_soc50)

DCIR 펄스 early  $\sqrt{t}$  외삽  $t \rightarrow 0$  절편. 순수 옴이 아니라 초기 비옴 성분 proxy. | 계산:  $R(t) = |\Delta V|/|I|$ ;  $t \in [0, 0.3]$ s 에서  $R$  vs  $\sqrt{t}$  회귀 절편. SOC50 블록  $\rightarrow$  forward-fill.

트렌드: 옴 저항 (SOC50): early=1.394  $\rightarrow$  late=2.374 ( $\Delta = +0.9802$ , +0.4981m $\Omega$ /100cyc)  $\rightarrow$  increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### 전하전달 저항 (SOC50) (R\_ct\_soc50)

중간 시간대 잔차의 지수 포화항. 계면/화학 저항 proxy (Cdl 직접 추정 아님). | 계산:  $\text{resid} = R - R_{\Omega} - A\sqrt{t}$ ;  $t \in [0.3, 10]$ s 에서  $R_{ct}(1 - e^{-(t/\tau)})$  fit.

트렌드: 전하전달 저항 (SOC50): early=0.7244  $\rightarrow$  late=0.9425 ( $\Delta = +0.2181$ , +0.09305m $\Omega$ /100cyc)  $\rightarrow$  increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### Rct 시정수 $\tau$ (tau\_ct\_soc50)

$R(t)$  잔차 fit의 유효 시정수.  $\tau \approx R_{ct} \cdot C_{dl}$  가설이지만 Cdl은 분리하지 않음. | 계산: curve\_fit 초기값 2s, bounds [0.05, 20]; 데이터로 추정.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

### 기계/화학 비 (mech\_vs\_chem\_ratio)

옴(스택·접촉) vs 전하전달(계면) 상대 비중. | 계산:  $R_{\text{ohmic\_soc50}} / R_{\text{ct\_soc50}}$ .

트렌드: 기계/화학 비: early=1.924  $\rightarrow$  late=2.519 ( $\Delta = +0.5948$ , +0.35921/100cyc)  $\rightarrow$  increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### R $\Omega$ 성장률 /100cyc (R\_ohmic\_growth\_100)

기준 DCIR 대비 100사이클당 옴 저항 증가율. 레벨( $R_{\Omega}$ )과 별개 — 감속해도  $R_{\Omega}$ 는 계속 상승 가능. | 계산:  $(R_{\Omega} - R_{\Omega 0}) / ((\text{cyc} - \text{cyc0})/100)$ .

트렌드: R $\Omega$  성장률 /100cyc: early=0.7957  $\rightarrow$  late=0.4668 ( $\Delta = -0.3289$ , -0.3314m $\Omega$ /100cyc/100cyc)  $\rightarrow$  decreasing · context | aging 기대=either | vs=context

### R $\Omega$ / R30s 분율 (R\_ohmic\_frac\_soc50)

30s 총저항 중 옴 성분 비중. | 계산:  $R_{\text{ohmic}} / R_{\text{30s\_total}}$  @ SOC50.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

## 지표 설명 · 트렌드 (M02Ch104)

### EoC 방전 10s DCIR (EoC\_dchgR\_10s)

충전 종료 후 방전 시작 10초 시점  $\Delta V/I$ . Rct·확산 포함 총 DCIR. | 계산:  $|V_0 - V(10s)|/|I| \cdot 1000$ . 순수 R $\Omega$  아님.

트렌드: EoC 방전 10s DCIR: early=0.1714 → late=0.2353 ( $\Delta$ =+0.06394, +0.02685m $\Omega$ /100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### EoC 방전 30s DCIR (EoC\_dchgR\_30s)

충전 후 방전 30초 시점 DCIR. | 계산:  $|V_0 - V(30s)|/|I| \cdot 1000$ .

트렌드: EoC 방전 30s DCIR: early=0.4605 → late=0.607 ( $\Delta$ =+0.1465, +0.06116m $\Omega$ /100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### EoC 방전 60s DCIR (EoC\_dchgR\_60s)

충전 후 방전 60초 시점 DCIR. | 계산:  $|V_0 - V(60s)|/|I| \cdot 1000$ .

트렌드: EoC 방전 60s DCIR: early=0.8075 → late=1.035 ( $\Delta$ =+0.2271, +0.09536m $\Omega$ /100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### EoD 충전 10s DCIR (EoD\_chgR\_10s)

방전 종료 후 충전 시작 10초 DCIR. | 계산:  $|V_0 - V(10s)|/|I| \cdot 1000$ .

트렌드: EoD 충전 10s DCIR: early=0.2743 → late=0.3472 ( $\Delta$ =+0.07291, +0.02702m $\Omega$ /100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### 저SOC 히스테리시스 (hyst\_area\_low)

저SOC 충·방전 전압 면적. Si-on-Gr chemo-mech 지표. | 계산: V-Q 히스테리시스를 SOC 밴드별로 적분 (low band).

트렌드: 저SOC 히스테리시스: early=0.07466 → late=0.04496 ( $\Delta$ =-0.0297, -0.01201V/100cyc) → decreasing · opposite\_aging | aging 기대=increase | vs=opposite\_aging

### 고SOC 히스테리시스 (hyst\_area\_high)

고SOC 히스테리시스. PE activity 보조 증거. | 계산: 히스테리시스 high-SOC band 면적.

트렌드: 고SOC 히스테리시스: early=0.2198 → late=0.2408 ( $\Delta$ =+0.02105, +0.008549V/100cyc) → flat · stable | aging 기대=increase | vs=stable

## 지표 설명 · 트렌드 (M02Ch104)

### 방전 플래토 폭 (dchg\_plateau\_width)

방전 플래토 구간 폭. 좁아지면 PE isolation/activity 패턴. | 계산: 방전 V 플래토 검출 후 Q 폭.

트렌드: 방전 플래토 폭: early=16.87 → late=14.8 ( $\Delta$ =-2.067, -1.075Q-units/100cyc) → decreasing · matches\_aging | aging  
기대=decrease | vs=matches\_aging

### 방전 플래토 $\Delta V$ (delta\_dchg\_plateau\_V)

기준 대비 방전 플래토 전압 이동. | 계산: plateau\_V(cycle) - plateau\_V(baseline).

트렌드: 방전 플래토  $\Delta V$ : early=-0.03799 → late=-0.08104 ( $\Delta$ =-0.04305, -0.01868V/100cyc) → decreasing · context | aging  
기대=either | vs=context

### 충전 dQ/dV 피크1 V (chg\_dQdV\_peak1\_V)

충전 IC 1번 피크 위치. FC-OCP 매칭·위상 이동 추적. | 계산: Q 보간 → SG → find\_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.741 → late=3.784 ( $\Delta$ =+0.043, +0.01064V/100cyc) → flat · context | aging  
기대=either | vs=context

### LAM 곡선 proxy (LAM\_curve\_proxy)

기준 V-Q 대비 scale 축소 proxy. 절대 LAM% 아님; 부호·bound 포화에 주의. | 계산: (1 - fit\_scale) \* 100 (discharge 3-param fit).  
scale>1 이면 음수.

트렌드: LAM 곡선 proxy: early=-2.983 → late=-16.28 ( $\Delta$ =-13.3, -5.282%/100cyc) → decreasing · context | aging  
기대=either | vs=context

### LLI 곡선 proxy (LLI\_curve\_proxy)

기준 대비 Q축 offset. 절대 LLI% 아님. | 계산: fit\_offset / Q\_max × 100.

트렌드: LLI 곡선 proxy: early=-0.6179 → late=-2.925 ( $\Delta$ =-2.307, -1.265%/100cyc) → decreasing · context | aging  
기대=either | vs=context

### Rate capability factor (RCF)

0.5C 용량 / 가까운 C/3 RPT 용량. rate 손실 시 하락. | 계산: Q\_0.5C(N) / Q\_C/3(nearest RPT).

트렌드: Rate capability factor: early=0.9711 → late=0.9569 ( $\Delta$ =-0.01422, -1.5e-061/100cyc) → flat · stable | aging  
기대=decrease | vs=stable



# 지표 설명 · 트렌드 (M02Ch104)

## 분극 효율 비 (PER)

과전위 대비 DCIR 스케일. 수송 제한 시 변화. | 계산:  $\eta(\text{SOC50}) / (\Delta I \cdot R_{\text{DCIR}_50})$ .

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

## 최대 과전위 $\eta$ (eta\_max)

C/3 vs 0.5C 동일 Q속 최대 전압차. | 계산: 동일 Q에서  $|V_{\text{C3}} - V_{\text{0.5C}}|$  최댓값.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

## $\eta$ 최댓값 SOC (eta\_argmax\_SOC)

$\eta$ 가 최대인 SOC. 고SOC면 PE, 저SOC면 NE 쪽 힌트. | 계산: argmax\_SOC of  $\eta(\text{SOC})$ .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

## PE activity 패턴 (LAM\_PE\_pattern\_score)

NCM 이차 고립/activity 패턴. 절대 LAM% 금지. | 계산: plateau·dQdV·LAM\_curve\_proxy·dQV\_log\_var 가중합 (mode\_weights).

트렌드: PE activity 패턴: early=0.6051 → late=0.9263 ( $\Delta=+0.3212$ ,  $+0.14480-1/100\text{cyc}$ ) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

## 접촉/스택 손실 (contact\_loss\_score)

옴 성장·분율·mech/chem 증가 증거 합. 전극 미분해. | 계산: R\_ohmic\_growth,  $\Delta R\Omega$ ,  $R\Omega_{\text{frac}}$ , EoC\_10s, mech/chem 가중합.

트렌드: 접촉/스택 손실: early=0.8664 → late=0.9423 ( $\Delta=+0.0759$ ,  $+0.04940-1/100\text{cyc}$ ) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

## LLI 패턴 (LLI\_pattern\_score)

CE·slippage·곡선 offset 기반 LLI 가설 점수. | 계산: CE ↓, CI ↑, restV/cutoff margin, LLI\_curve\_proxy 등.

트렌드: LLI 패턴: early=0.3694 → late=0.6369 ( $\Delta=+0.2675$ ,  $+0.12010-1/100\text{cyc}$ ) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

## 지표 설명 · 트렌드 (M02Ch104)

### 계면 R 패턴 (interface\_R\_score)

Rct· $\tau$ ·VE 하락 등 계면저항 성장 가설. | 계산: R\_ct ↑, tau\_ct ↑, VE ↓, EoC\_60s 등.

트렌드: 계면 R 패턴: early=0.4752 → late=0.8706 ( $\Delta$ =+0.3954, +0.14840-1/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### 고체확산 제한 (solid\_diffusion\_score)

A\_diff·PER·RCF 기반 확산/수송 제한. | 계산: A\_diff ↑, PER ↑, RCF ↓ 등.

트렌드: 고체확산 제한: early=0.5208 → late=0.6973 ( $\Delta$ =+0.1764, +0.026140-1/100cyc) → flat · stable | aging 기대=increase | vs=stable

### PE lean 점수 (PE\_side\_score)

LAM\_PE\_pattern + feature boost + FC-OCP  $\Delta$ hits. | 계산: 0.75 · LAM\_PE + 0.20 · PE\_boost + peak\_boost.

트렌드: PE lean 점수: early=0.5038 → late=0.7947 ( $\Delta$ =+0.2909, +0.12810-1/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### contact\_stack lean (contact\_stack\_score)

contact\_loss를 dominant 경쟁에 넣는 이름.  $\approx$  contact\_loss\_score. | 계산: clip(contact\_loss\_score, 0, 1).

트렌드: contact\_stack lean: early=0.8664 → late=0.9423 ( $\Delta$ =+0.0759, +0.04940-1/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### NE 가설 점수 (NE\_side\_score)

contact × Si chemo-mech co-sign. Si 없으면 NE 확정 금지. | 계산: 0.55 · contact · (0.25+0.75 · si) + 0.30 · NE\_boost · si.

트렌드: NE 가설 점수: early=0.1921 → late=0.285 ( $\Delta$ =+0.09298, +0.039660-1/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging

### Si co-sign (si\_cosign)

저SOC hyst·Q\_relax·mech/chem·CV 등 Si chemo-mech 동시 신호. | 계산: SI\_NE\_COSIGN feature boost (baseline 대비 ↑ 비율).

트렌드: Si co-sign: early=0.2 → late=0.4 ( $\Delta$ =+0.2, +0.070320-1/100cyc) → increasing · matches\_aging | aging 기대=increase | vs=matches\_aging